

RECONSTRUCTING SOUND FROM EEG USING LARGE-LANGUAGE MODELS

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Current **neural decoding** methods reconstruct specific acoustic features from EEG using either linear approaches (e.g., envelope tracking) or deep learning models (CNNs, RNNs for spectrogram prediction). However, both type of approaches remain **limited to specific acoustic components** (e.g., the envelope) and cannot reconstruct the **complete sound signal** with all its complexity.

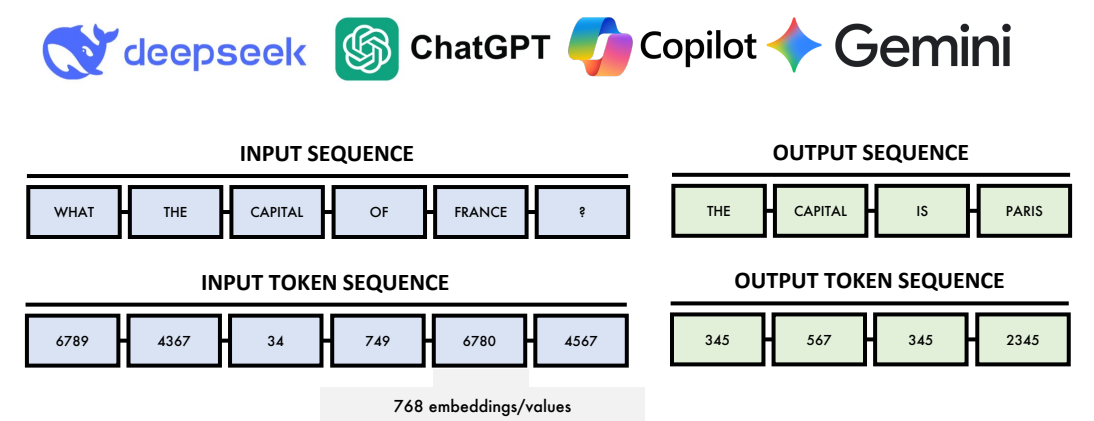
Here, we introduce a novel framework that leverages **Large Language Model (LLM)** architectures. Specifically, we leverage a model – **NeuroLM** – a GPT-2-based model pre-trained on large-scale EEG datasets – to achieve full sound reconstruction from brain activity.

LLMs might be a good candidate for such task due to their capacity to handle **continuous sequential data** and capture long-range temporal dependencies.

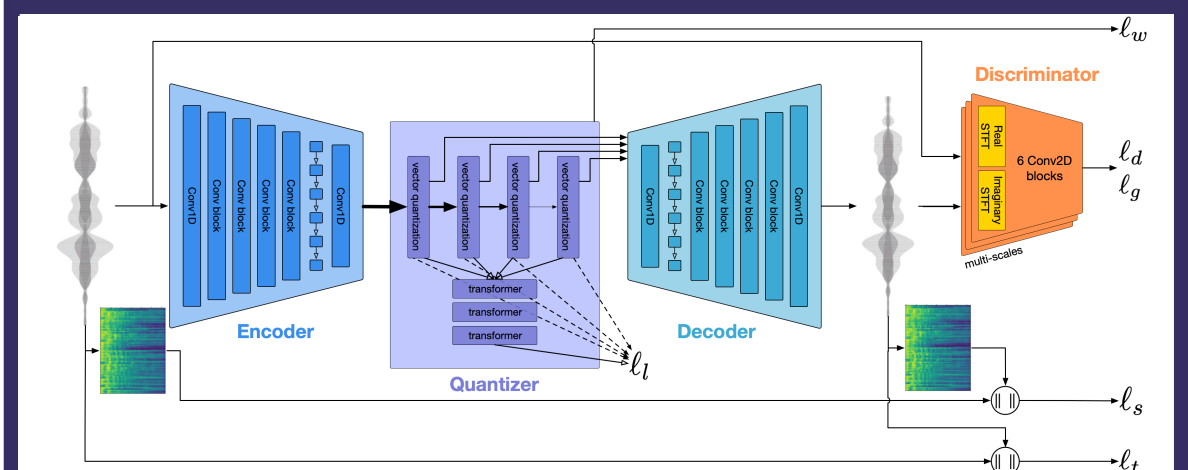
To our knowledge, this represents the first application of LLMs for complete sound reconstruction from neural signals, enabling direct transformation of EEG sequences into acoustic tokens and subsequently into intelligible, continuous audio.

Beyond its methodological innovation, this approach provides a powerful clinical tool for audiology research and hearing diagnostics, enabling objective assessment of how the brain encodes sound in both typical and impaired hearing.

LLM AND GPT STRUCTURE

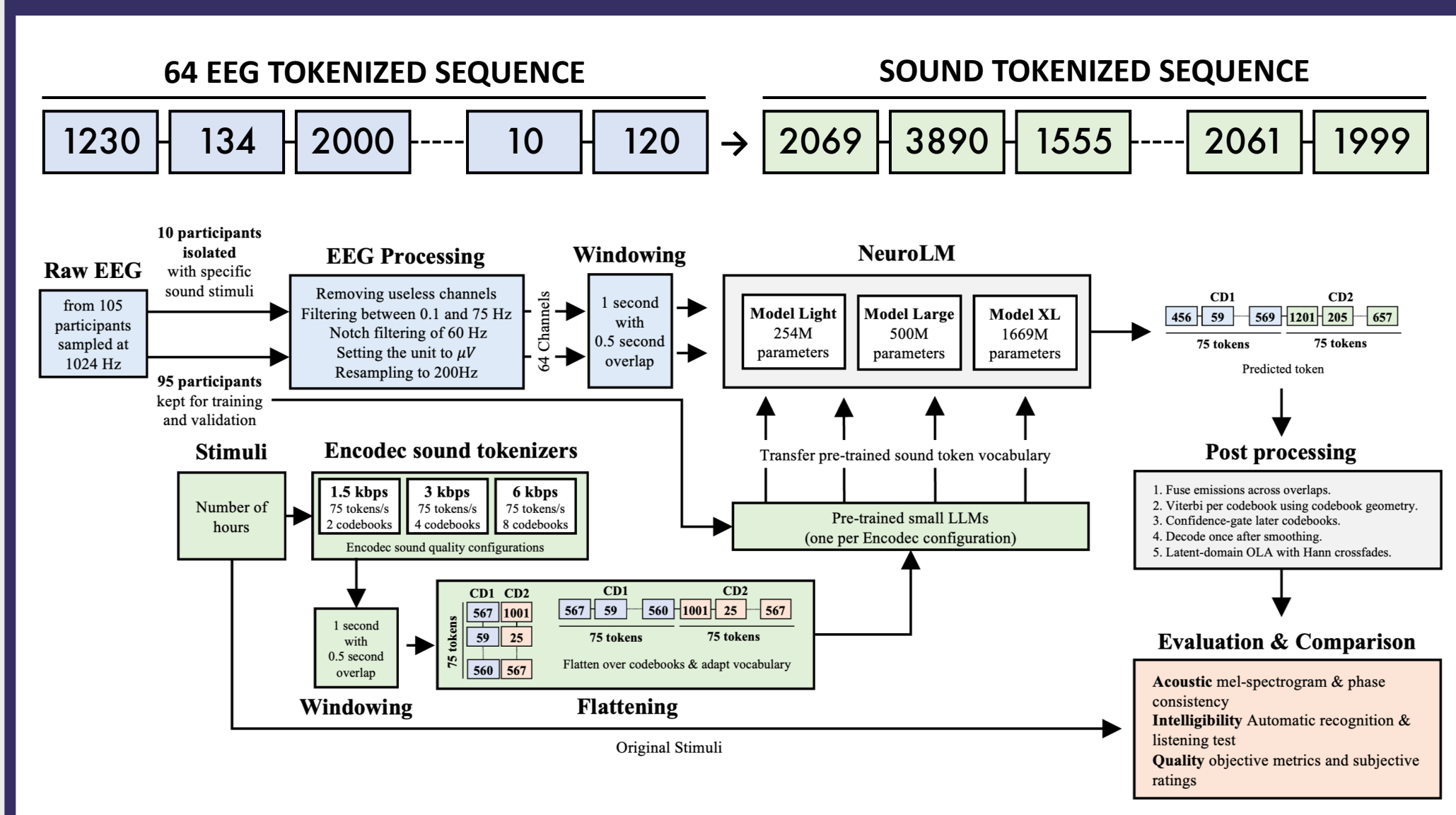


SOUND TOKENIZATION

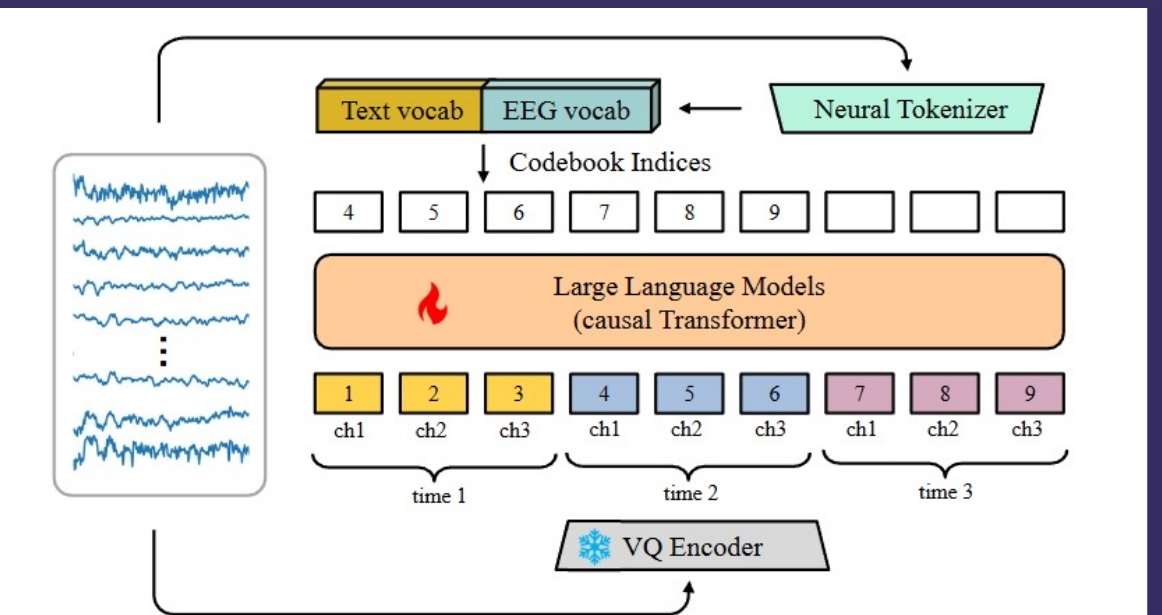


We use EnCodec, a neural audio codec developed by Meta AI, to convert continuous audio into discrete acoustic tokens. EnCodec employs residual vector quantization (RVQ) to compress sound into multiple hierarchical codebooks (1.5-6 kbps bitrates), where each codebook captures different levels of acoustic detail - from coarse spectral structure to fine temporal dynamics. This creates a vocabulary of acoustic patterns analogous to words in language. Each audio segment is represented as a sequence of tokens across multiple codebooks, enabling the language model to learn and generate speech by predicting these discrete acoustic units rather than continuous waveforms.

OVERVIEW OF OUR APPROACH



EEG TOKENIZATION

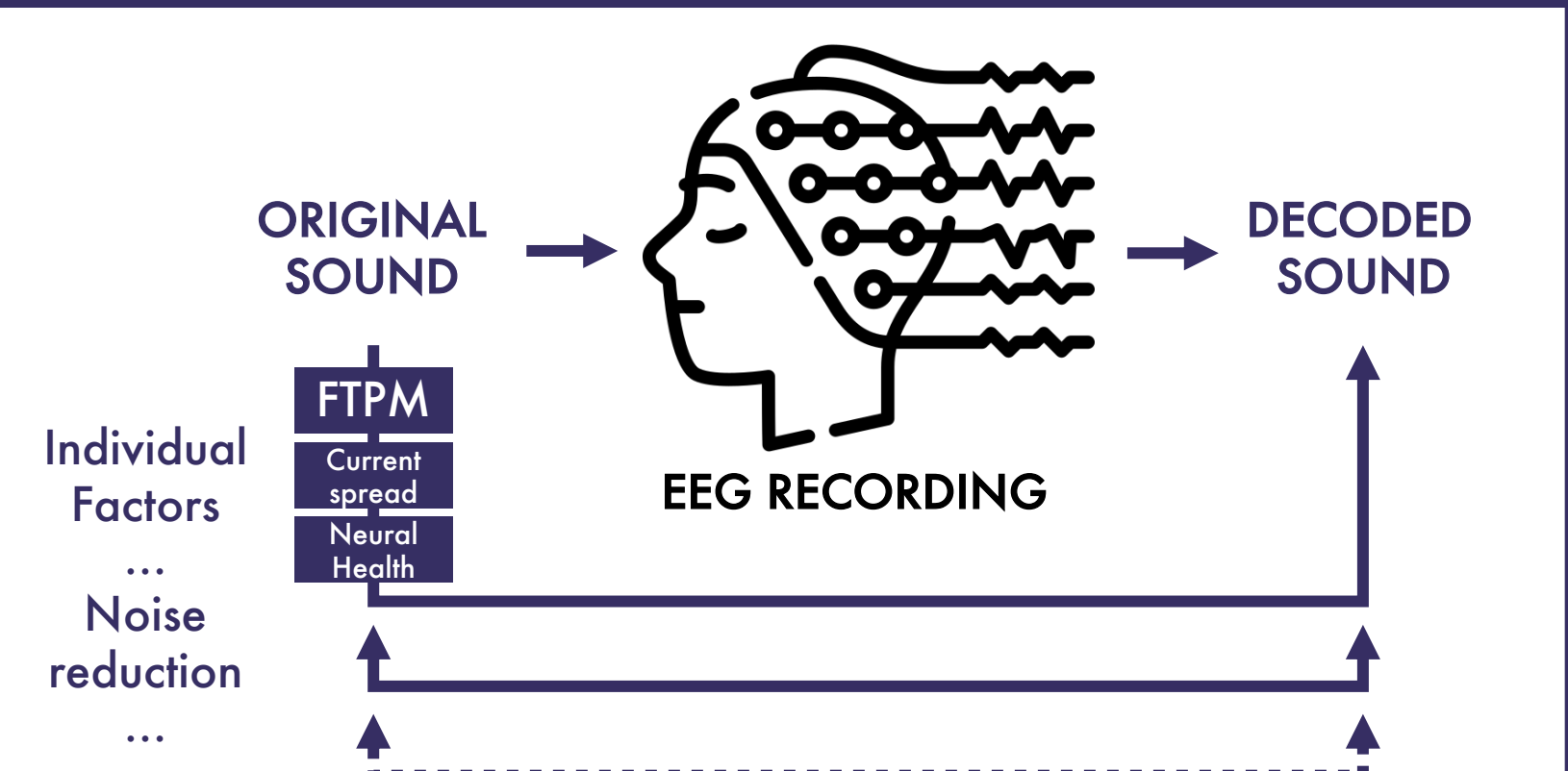


NeuroLM is a multi-task model for EEG analysis. It treats brain signals as a "foreign language", encoding raw EEG into discrete neural tokens processed by a LLM pre-trained on 25,000 hours of data.

PRELIMINARY CONCLUSION

- ✅ **Framework Established:**
 - First LLM-based approach for complete sound reconstruction from EEG
 - Scalable architecture validated across three model sizes
 - End-to-end neural-to-acoustic tokenization pipeline functional
 - Proof-of-concept for autoregressive speech generation from brain signals
- ⚠️ **Current Challenge: Ensuring EEG-Driven Predictions**
 - Model demonstrates strong autoregressive capabilities (i.e., learns audio token patterns) **BUT may underutilize EEG signals during inference**
 - Risk: Predictions rely on learned audio priors rather than neural activity
 - Common issue in multimodal learning (models take "shortcut" through easier signal)
- 🔧 **Active Intervention:**
 - Retraining with **EEG-only context** (removing audio token history)
 - Forces model to predict **solely from neural input**

CLINICAL PERSPECTIVES



IMMEDIATE NEXT STEP

Completing an EEG-only retraining to validate neural-driven predictions.
Continue the training.

ALTERNATIVE APPROACH

Predicting words or phonemes instead of acoustic tokens.

This strategy offers several advantages:

- smaller vocabulary** (40-50 phonemes versus 1024*8 acoustic combinations),
- the smaller sequence** per sample (few words or phonemes instead of hundreds of audio tokens).

This alternative approach would be slightly different and conduct to different clinical interpretations.